

Dennies Bor

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EDUCATION

- **George Mason University** Fairfax, VA, USA
PhD in Earth Systems and Geoinformation Sciences (Advisor: Dr. Edward Oughton) Sep 2023 – Present

Expected graduation: May 2027
Selected Coursework: Quantitative Methods; Remote Sensing; Hyperspectral Imaging; Earth Image Processing; Spatial Computing; Applied Electromagnetic Theory; Computational Physics II; Digital Signal Processing; Earth Science Data / Advanced Data Analysis.
GPA: 3.85
- **George Mason University** Fairfax, VA, USA
MS in Geoinformatics and Geospatial Intelligence May 2026 – Dec 2026
- **Technical University of Kenya** Nairobi, Kenya
BEng in Aeronautical Engineering, First Class Honors Sep 2013 – May 2019

RESEARCH & PROFESSIONAL EXPERIENCE

- **Graduate Research Assistant** George Mason University, VA, USA
Space Weather, Computational Modeling, and Infrastructure Resilience May 2022 – Present
 - Develop coupled hazard and impact models integrating physical simulation, geospatial processing, infrastructure vulnerability, and socioeconomic impact estimation.
 - Develop the Coupled Space Weather Impact Model (C-SWIM) to assess space weather impacts on the U.S. power grid, satellite networks, and the wider economy.
 - Develop and maintain HAPI server software under the supervision of Dr. Robert S. Weigel.
 - Conduct numerical simulations, uncertainty quantification, and national scale infrastructure analyses using local and cloud computing workflows.
 - Build reproducible research software for data ingestion, scenario generation, simulation, postprocessing, visualization, and dissemination of research results.
- **NSF NCAR Supported Research** George Mason University, VA, USA
Multi-Hazard Transmission Network Risk Assessment (MHTRAN) Jan 2026 – May 2026
 - Undertook Curricular Practical Training (CPT) supported by the NSF National Center for Atmospheric Research (NSF NCAR) to develop MHTRAN, a framework for assessing the impacts of multiple natural hazards on critical infrastructure.
 - Generalized methods developed through C-SWIM to assess and compare the vulnerability of the U.S. high voltage transmission network to multiple natural hazards, including space weather.
 - Integrated national hazard datasets, infrastructure fragility models, population exposure, and economic impact models within a common computational framework.
- **Research Intern** Syllab Systems, Texas, USA
Spacecraft Cyber Resilience and Space Weather Summer 2026
 - Develop artificial intelligence capabilities for the STAR platform to improve spacecraft cyber resilience and mission readiness.
 - Combine synthetic spacecraft telemetry, space weather modeling, machine learning detection, and agent based adjudication to distinguish cyber threats from natural environmental effects.
 - Integrate C-SWIM space weather models with satellite telemetry and cyber threat analysis.
 - Build supporting data, application programming interface (API), and cloud infrastructure for traceable decision support for spacecraft operators.

- Supported satellite tracking and telemetry for geospatial applications.
- Assisted in the maintenance and operation of radio frequency (RF) communication systems.

PUBLICATIONS & PREPRINTS

*Equal contribution.

- **Major Space Weather Risks Identified via Coupled Physics–Engineering–Economic Modeling:** Oughton, E. J. *, **Bor, D. K. ***, Weigel, R. S., Gaunt, C. T., Dogan, R., Huang, L., Love, J. J., Wiltberger, M. *AGU Advances*, 7(6), e2026AV002367, 2026. doi:10.1029/2026AV002367 — arXiv:2412.18032 — Code: C-SWIM
- **Systematic Component-Level Characterization of Electricity Transmission Infrastructure for Space Weather Risk Assessment:** **Bor, D. K. ***, Oughton, E. J. *, Peters, E. A. *, Rivera, N., Gaunt, C. T., Weigel, R. S., Wiltberger, M. *Space Weather*, 24(9), e2026SW005166, 2026. doi:10.1029/2026SW005166 — arXiv:2412.17685 — Code: C-MAP
- **GIC-Related Observations During the May 2024 Geomagnetic Storm in the United States:** Wilkerson, L. A., Weigel, R. S., Thomas, D., **Bor, D.**, Oughton, E. J., Gaunt, C. T., Balch, C. C., Wiltberger, M. J., Pulkkinen, A. *Space Weather*, 24(5), e2025SW004758, 2026. doi:10.1029/2025SW004758 — arXiv:2507.07009
- **Assessing the Economic Benefits of Space Weather Mitigation Investment Decisions: Evidence from Aotearoa New Zealand:** Oughton, E. J., Renton, A., Mac Manus, D. H., **Bor, D.**, Rodger, C. J. *Space Weather*, in press, 2026. doi:10.1029/2025SW004919 — arXiv:2507.12495
- **C-SWIM: A Coupled Space Weather Impact Model for Satellite Fleet Vulnerability and Economic Loss Under a 1-in-100-Year Solar Energetic Particle Event:** **Bor, D.**, Oughton, E. J., Weigel, R. S., Yang, R., Clower, T., Wiltberger, M. J., Linares, R. Under review, 2026. arXiv:2605.22576 — Code: C-SWIMs
- **A Comparative Multi-Hazard Risk Assessment of the US High-Voltage Transmission Network:** **Bor, D.**, Oughton, E. J., Weigel, R. S., Yang, R., Clower, T., Newman, A., Valle, A. R. Preprint, 2026. arXiv:2605.23053 — Code: MHTRAN
- **Comparing Object Detection Models for Electrical Substation Component Mapping:** Mody, H., Bansal, N., **Bor, D. K.**, Oughton, E. J. Preprint, 2025. arXiv:2512.22454

RESEARCH SOFTWARE & PROJECTS

- **Coupled Space Weather Impact Model (C-SWIM):** Physics, engineering, and economic modeling framework for assessing the impacts of space weather on critical infrastructure. GitHub — Results Dashboard
- **C-SWIM for Satellite Fleet Vulnerability:** Framework for assessing solar energetic particle exposure, satellite vulnerability, and economic impacts across large satellite fleets. GitHub
- **C-SWIM Mapping and Annotation Platform (C-MAP):** Web based platform for mapping and characterizing electricity transmission infrastructure using satellite and aerial imagery. GitHub
- **Multi-Hazard Transmission Network Risk Assessment (MHTRAN):** National scale framework for comparing the impacts of natural hazards on the U.S. high voltage transmission network. GitHub
- **GIC Prediction Model Evaluation (tfpy):** GitHub
- **Computer Vision for Power Substation Asset Detection:** GitHub

RESEARCH FUNDING

- **NSF National Center for Atmospheric Research, Supplemental Research Support, 2026:** Supported development of the Multi-Hazard Transmission Network Risk Assessment framework to generalize methods developed through C-SWIM to other natural hazards affecting critical infrastructure.

RESEARCH MENTORING

- **Aspiring Scientists Summer Internship Program (ASSIP), George Mason University:** Mentored more than five high school students in research involving scientific computing, geospatial analysis, space weather, and infrastructure resilience.

PROGRAMS & AFFILIATIONS

- **National Center for Atmospheric Research** USA
Early Career Faculty Innovators Program 2023 – 2025
- **African Institute of Mathematical Sciences** South Africa
Africa Data Science Intensive Program 2022

TECHNICAL SKILLS

- **Coupled Systems Modeling:** Physics based modeling, power grid risk modeling, satellite risk modeling, and socioeconomic impact modeling using input-output methods.
- **Uncertainty Quantification:** Probabilistic simulation, Monte Carlo methods, scenario design, sensitivity analysis, and validation workflows.
- **Geospatial & Remote Sensing:** Infrastructure mapping, GIS algorithms, satellite and aerial imagery processing, and reproducible spatial analysis.
- **Scientific Computing:** Python scientific computing, numerical methods, optimization using Pyomo and IPOPT, and scalable scenario studies.
- **Research Software:** Open source research software, databases, application programming interfaces, dashboards, cloud computing, and version controlled research pipelines.

REFERENCES

Available upon request.